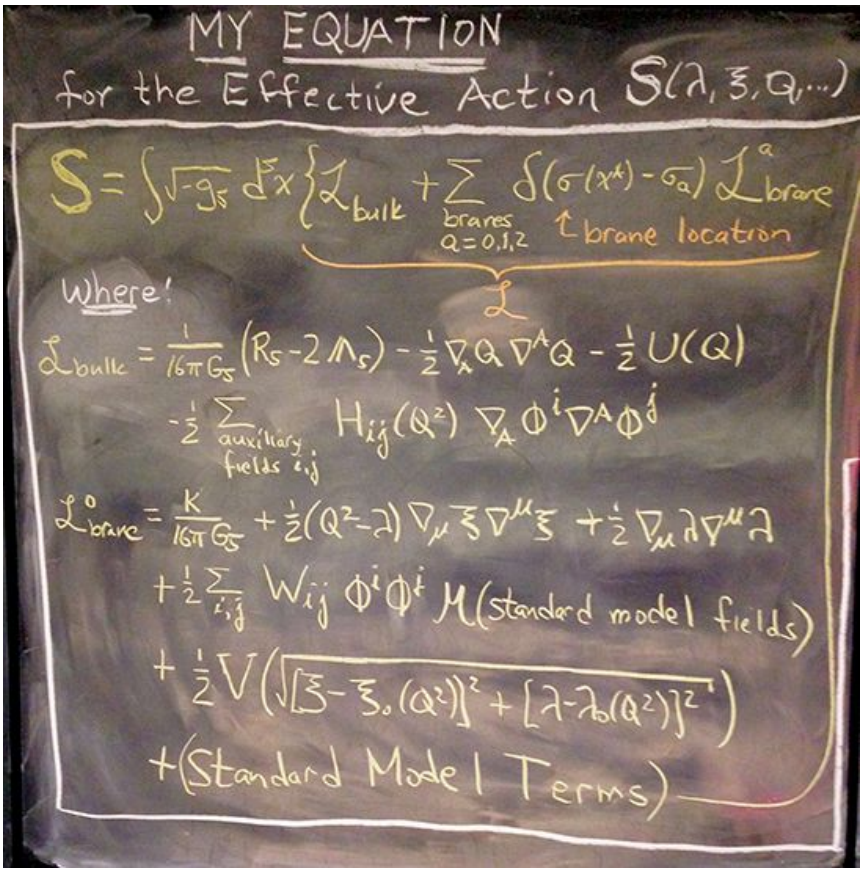


Fig. 25.7. Professor Brand's equation.

This equation is called an "Action." There is a well-known (to physicists) mathematical procedure to begin with such an Action, and from it deduce *all* the nonquantum physical laws. The Professor's equation, in effect, is the mother of all nonquantum laws. But for it to give birth to the *right* laws—the laws that predict *correctly* how the anomalies are produced, how the wormhole is held open, how G is controlled, and how our universe is protected—the equation must have precisely the correct mathematical form. The Professor doesn't know the correct form. He is guessing. His is an educated guess, but a guess nevertheless.

His equation contains lots of guessing: guesses for the things called "$U(Q)$, $H_{ij}(Q^2)$, W_{ij}, and $\mathcal{M}$(standard model fields)" on his blackboard (Figure 25.7). In effect, these are guesses for the nature of the bulk